

Spectral Equivalence of Solvable Lie Algebras with Nilradical I: General and Heisenberg Cases

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Abstract

In recent years, the characteristic polynomial of finite-dimensional Lie algebras has been proposed as a tool for a spectral classification of solvable algebras. We resolve two questions of Key and Yang (2021) in this direction. We show that two spectral invariants, $k(\mathfrak{s})$ and the higher Betti numbers of the eigenvariety complement, are determined by the weight space decomposition of the nilradical under the action of its extension. The second result is obtained by identifying the characteristic polynomial with a matroid and computing the cohomology ring via the associated Orlik-Solomon algebra. As an application, we obtain a complete spectral classification of all solvable Lie algebras with three- and five-dimensional Heisenberg nilradical.

1 Introduction

1.1 Solvable Lie Algebras and Nilradicals

Throughout this paper, we work with finite-dimensional Lie algebras over the complex field $\mathbb{C}$. Classifying these structures is one of the oldest and most important problems in Lie theory. While the classification of semisimple Lie algebras has been essentially solved for over a century through the works of Killing and Cartan [Kil90, Car94], the solvable case remains an open problem.

Recall the decomposition theorems of Levi [Lev05] and Malcev [Mal45]. Levi's Theorem states that every finite-dimensional Lie algebra $\mathfrak{g}$ decomposes as a semidirect product $\mathfrak{g} = \mathfrak{s} \ltimes \text{Rad}(\mathfrak{g})$, where the radical $\text{Rad}(\mathfrak{g})$ is the unique maximal solvable ideal and $\mathfrak{s}$ is a semisimple subalgebra. Malcev adds that the Levi factor $\mathfrak{s}$ is unique up to inner automorphism. Together, these results reduce the classification of finite-dimensional Lie algebras to the study of solvable Lie algebras (the radical itself) and the actions of semisimple Lie algebras on them. Of these tasks, the classification of solvable Lie algebras is substantially more challenging. In general, a complete classification cannot be expected. Consequently, the main obstruction to a full classification of finite-dimensional Lie algebras lies in the structure of the solvable radical $\text{Rad}(\mathfrak{g})$.

Since the radical $\text{Rad}(\mathfrak{g})$ is itself a solvable Lie algebra, its structure is primarily determined by its maximal nilpotent ideal, the nilradical. Every solvable Lie algebra $\mathfrak{g}$ contains a unique maximal nilpotent ideal $\text{Nil}(\mathfrak{g})$; if $\mathfrak{g}$ is not nilpotent, this ideal is non-trivial and contains the derived algebra $[\mathfrak{g}, \mathfrak{g}]$. Consequently, we may study $\mathfrak{g}$ through the vector space decomposition $\mathfrak{g} = \text{Nil}(\mathfrak{g}) \oplus \mathfrak{f}$, where $\mathfrak{f}$ is a complementary subspace acting on $\text{Nil}(\mathfrak{g})$ via outer derivations. This idea motivates the classification strategy initiated by Mubarakzhanov [Mub63c], which proceeds to classify solvable Lie algebras by first fixing a nilpotent Lie algebra and then determining all of the possible extensions that produce a solvable Lie algebra.

Historically, classification efforts have proceeded by dimension. Complete lists of nilpotent Lie algebras are available up to dimension seven [Mor58, Saf64, See93, Gon98], while computer-assisted approaches have given some results for dimensions eight and nine [Tsa99, TKK00]. In the solvable case, the classification is also complete up to dimension seven. The classification up to dimension six was completed decades ago by [Mub63c, Mub63a, Mub63b, Tur90], with [SW14] providing an exposition of these results. The classification in dimension seven was finished recently by assembling partial results organized according to the dimension of the nilradical: the last remaining cases where the included codimension-one nilradicals [Par07], the four-dimensional case [HT08], and the five-dimensional case [LNN⁺21].

Following the method introduced by Mubarakzhanov, mathematicians have made substantial progress in classifying solvable Lie algebras with specific nilradicals. In particular, this approach has been successfully

applied to solvable Lie algebras with abelian nilradical [NW94], triangular nilradical [TW98], and Borel nilradicals [SW12]. Moreover, there has been plenty of work on solvable Lie algebras with filiform and quasifiliform nilradical [SW05, SW09, ACSGV06, WLD08, SK10, ABCSGV11]. It is also well established that some nilpotent Lie algebras, called characteristically nilpotent, cannot be the nilradical of any solvable Lie algebra and should be excluded from the consideration [Kha90, GH94].

This series of papers builds on the spectral approach to finite-dimensional Lie algebras introduced in [HZ19, KY21]. Our goal is to understand the spectral theory of solvable Lie algebras through the structure of their nilradicals. In this paper, we study the general theory and apply it to solvable Lie algebras with Heisenberg nilradical. Subsequent papers extend this framework to solvable Lie algebras with Borel and triangular nilradicals, and finally to those with filiform and quasifiliform nilradicals.

Among the non-abelian Lie algebras, the Heisenberg Lie algebra $\mathfrak{h}(m)$ stands out as one of the simplest and most important examples of a nilpotent Lie algebra. It arises naturally in several mathematical and physical contexts. In quantum mechanics, it gives the canonical commutation relations between position and momentum operators; in symplectic geometry, it appears as the Lie algebra of the Heisenberg group, which acts transitively on the phase space; and in representation theory, it serves as an important example of a non-semisimple Lie algebra with rich representation theory. For a comprehensive treatment of these connections, we refer the interested reader to [Fol89, Kir04, Woi24].

Consequently, a spectral classification of solvable Lie algebras with Heisenberg nilradical is a natural step in the development of spectral theory for finite-dimensional Lie algebras. Physically, these Lie algebras classify the admissible dynamical symmetries of non-linear Schrödinger-type equations [GW88] and constitute the solvable subalgebras of the symplectic nuclear collective motion model $\mathfrak{sp}(2n, \mathbb{R}) \ltimes \mathfrak{h}(n)$ [RR77]. In the context of invariant theory, they require the use of rational Casimir operators to distinguish coadjoint orbits where polynomial invariants fail [PSWZ76]. Geometrically, these algebras can be used to construct noncompact Einstein solvmanifolds and Ricci solitons [Heb98, Lau10].

1.2 Characteristic Polynomials, Spectral Equivalence, and Spectral Invariants

The spectral theory of a single linear operator $A \in M_k(\mathbb{C})$ is traditionally developed via its characteristic polynomial $Q_A(\lambda) = \det(\lambda I - A)$. This invariant gives the eigenvalues and algebraic multiplicities that determine the operator's structure. However, algebraic structures arising from physics and geometry are rarely generated by a single operator, but rather by sets of non-commuting operators $\{A_1, \dots, A_n\}$. Developing a spectral theory for such sets gives an immediate obstruction, since non-commutativity prevents one from simultaneously diagonalizing the generators or placing their spectral data into a single polynomial invariant using standard techniques.

The classical forerunner of a multivariate spectral theory is the theory of the group determinant, introduced by Dedekind and Frobenius in the 1890s. Although arising in a purely algebraic context, this construction displays the important features of a multivariate spectral invariant.

Let $G = \{1, g_1, \dots, g_n\}$ be a finite group with representation π . Recall that the group determinant is defined as:

$$Q_\pi(z) := \det(z_0 I + z_1 \pi(g_1) + \dots + z_n \pi(g_n))$$

Frobenius's breakthrough was the discovery that for the left regular representation, the factorization of this polynomial into irreducible factors completely determines the representation theory of the group.

Theorem 1.1 (Dedekind-Frobenius, [Ded69, Fro96]). *If $G = \{1, g_1, \dots, g_n\}$ is a finite group and λ_G is the left regular representation, then*

$$Q_{\lambda_G}(z) = \prod_{\pi \in \widehat{G}} Q_\pi(z)^{d_\pi},$$

where $\widehat{G}$ is the set of equivalence classes of irreducible unitary representations of G (called the unitary dual of G), and d_π is the dimension of the representation π . Moreover, each such Q_π is an irreducible polynomial.

This theorem marks the start of the study of representation theory for finite groups. For historical developments, see [Cur92, Cur03, Dic21, Dic02, FS91]. Despite this early success, it took more than a century before the group determinant was extended to other algebraic settings. We briefly outline this history; a comprehensive survey is available in [Yan24]. In 2009, [Yan09] extended this notion to unital

Banach algebras, which sparked a lot of work [BCY11, CY13, CSZ16, DY18, GY20, GY17, HWY17, MQW17, SYZ11]. Motivated by these results, [HZ19] defined the characteristic polynomial for finite-dimensional Lie algebras.

Definition 1.2 (Characteristic Polynomial). *Let L be a finite-dimensional Lie algebra with basis $\{x_1, \dots, x_n\}$. The **characteristic polynomial** of L is*

$$Q_L(z_0, \mathbf{z}) := \det \left(z_0 I + \sum_{i=1}^n z_i \operatorname{ad} x_i \right),$$

where $\operatorname{ad} x_i$ is the adjoint representation of x_i and $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{C}^n$.

This spectral theory has already been applied to study simple [AM24], classical [FLW26], and solvable Lie algebras [HZ19, Mul23]. The solvable case is particularly interesting: by Lie's theorem, the adjoint operators can be simultaneously upper triangularized over $\mathbb{C}$, and therefore the characteristic polynomial factors completely into linear forms.

Theorem 1.3 ([HZ19]). *Let L be a solvable Lie algebra over $\mathbb{C}$. Then the characteristic polynomial $Q_L(z_0, \mathbf{z})$ factors as*

$$Q_L(z_0, \mathbf{z}) = \prod_{j=1}^{\dim L} (z_0 + \ell_j(\mathbf{z})),$$

where each ℓ_j is a linear form in the variables $\mathbf{z} = (z_1, \dots, z_n)$.

One motivation for developing a theory of characteristic polynomials for Lie algebras is the expectation that these polynomials will give spectral invariants capable of distinguishing and classifying Lie algebras. Observe that if two Lie algebras L_1 and L_2 are isomorphic, any isomorphism corresponds to a change of basis matrix $B \in \operatorname{GL}_n(\mathbb{C})$. A direct calculation shows that their characteristic polynomials must satisfy $Q_{L_2}(z_0, \mathbf{z}) = Q_{L_1}(z_0, \mathbf{z}B)$. Motivated by this observation, [KY21] introduced a coarser classification criterion:

Definition 1.4 (Spectral Equivalence). *Two n -dimensional solvable Lie algebras L_1 and L_2 are **spectrally equivalent** over $\mathbb{C}$, denoted $L_1 \sim L_2$, if their characteristic polynomials satisfy*

$$Q_{L_2}(z_0, \mathbf{z}) = Q_{L_1}(z_0, \mathbf{z}B)$$

for some $B \in \operatorname{GL}_n(\mathbb{C})$.

Spectral equivalence is strictly weaker than isomorphism: if $L_1 \cong L_2$, then $L_1 \sim L_2$, but the converse fails in general. The strongest result in this direction is the following result for nilpotent Lie algebras:

Theorem 1.5 ([KY21]). *A Lie algebra L is nilpotent if and only if its characteristic polynomial is $Q_L(z_0, \mathbf{z}) = z_0^{\dim L}$.*

This does not extend to the broader class of solvable Lie algebras in a simple way. Nevertheless, the factorization from Theorem 1.3 still gives rise to an interesting invariant $k(L)$, introduced in [KY21], which can distinguish some non-isomorphic solvable Lie algebras.

Definition 1.6 (The Invariant $k(L)$). *Let L be a solvable Lie algebra. The integer $k(L)$ is defined as the number of distinct irreducible linear factors in the factorization of $Q_L(z_0, \mathbf{z})$.*

However, this invariant remains quite mysterious. This motivates the following problem:

Problem 1.7 ([KY21]). *Can the spectral invariant $k(L)$ be expressed in terms of other Lie-theoretic invariants?*

To date, progress on this question has been limited. The strongest available result establishes an upper bound for $k(L)$:

Proposition 1.8 ([KY21]). *Let L be an n -dimensional solvable Lie algebra with basis $\{x_1, x_2, \dots, x_n\}$. Then*

$$k(L) \leq \max_{1 \leq i \leq n} \{|\sigma(\operatorname{ad} x_i)|\}$$

where $\sigma(\operatorname{ad} x_i)$ denotes the spectrum of the adjoint operator $\operatorname{ad} x_i$, and $|\cdot|$ denotes the cardinality of the set.

We resolve this problem in Section 2.1. Theorem 2.2 establishes that $k(L)$ is completely determined by the weight space decomposition of the nilradical, specifically as the cardinality of the union of the weight set and the zero vector. Furthermore, Theorem 2.4 provides sharp lower and upper bounds for this invariant in terms of the dimensions of the extension and the nilradical, respectively.

Additionally, [KY21] showed that the characteristic polynomial admits several natural geometric interpretations, which in turn give rise to topological spectral invariants.

Definition 1.9 (Eigen-variety). *The **eigen-variety** of $\mathcal{L}$ is the zero locus of the characteristic polynomial in affine space $\mathbb{C}^{n+1}$:*

$$V_{\mathcal{L}} := \{(z_0, \mathbf{z}) \in \mathbb{C}^{n+1} \mid Q_{\mathcal{L}}(z_0, \mathbf{z}) = 0\}.$$

Since z_0 is always a factor of $Q_{\mathcal{L}}(z_0, \mathbf{z})$, the hyperplane $\{z_0 = 0\}$ is always present in $V_{\mathcal{L}}$. To study the non-trivial topology, we assume $Q_{\mathcal{L}}(z_0, \mathbf{z}) = z_0^k p(z_0, \mathbf{z})$, where z_0 is not a factor of the polynomial $p(z_0, \mathbf{z})$. We consider the variety $V_{\mathcal{L}}^* := \{(z_0, \mathbf{z}) \in \mathbb{C}^{n+1} \mid p(z_0, \mathbf{z}) = 0\}$.

In the case where $\mathcal{L}$ is solvable, since $Q_{\mathcal{L}}(z)$ factors into linear forms, $V_{\mathcal{L}}^*$ is a hyperplane arrangement. [KY21] proposed that studying the topology of the complement of the variety, $(V_{\mathcal{L}}^*)^c := \mathbb{C}^n \setminus V_{\mathcal{L}}^*$ would lead to topological spectral invariants. Therefore, they introduced the following definition:

Definition 1.10 (Poincaré Polynomial). *For a finite-dimensional Lie algebra L , the **Poincaré polynomial** is defined as*

$$P_L(t) := \sum_{j=0}^{\infty} b_j(L) t^j,$$

where $b_j(L) := \dim H^j((V_L^*)^c, \mathbb{C})$ for $j \geq 0$ is the j th **Betti number**.

Additionally, they proved two results:

Proposition 1.11 ([KY21]). *If L is a solvable Lie algebra, then $b_1(L) = k(L) - 1$.*

Proposition 1.12 ([KY21]). *For any finite-dimensional solvable Lie algebra L , we have $\deg P_L \leq \dim L / \text{nil}(L) + 1$*

However, the structure of the higher Betti numbers is still an open problem:

Problem 1.13 ([KY21]). *Can the Betti numbers $b_j(L)$, $j \geq 2$ be determined by known invariants of L ?*

We resolve this problem in Section 2.2. By identifying the cohomology of the eigenvariety complement with the Orlik–Solomon algebra of a matroid, Theorem 2.20 shows that the Poincaré polynomial is determined by the Whitney numbers of the first kind. This gives an explicit combinatorial formula for the Betti numbers as alternating sums of these Whitney numbers. Additionally, Theorem 2.22 determines the exact degree of the Poincaré polynomial as the dimension of the subspace spanned by the weights, while Theorem 2.29 establishes the log-concavity of the Betti numbers.

Finally, in section 3, we apply this general framework to the important class of solvable Lie algebras with Heisenberg nilradical $\mathfrak{h}(m)$. We derive a canonical factorization of the characteristic polynomial that explicitly isolates the symplectic structure of the nilradical (Theorem 3.3). This factorization reveals that the symplectic geometry imposes strict constraints on the topology of the eigenvariety complement, forcing the Betti numbers $b_j(\mathfrak{s})$ to vanish for $j > m + 1$ (Theorem 3.7). Finally, we provide a complete spectral classification and explicit computation of spectral invariants for all solvable extensions of $\mathfrak{h}(1)$ and $\mathfrak{h}(2)$.

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2 The General Case

2.1 The Spectral Invariant $k(L)$

Let $\mathfrak{s}$ be a finite-dimensional solvable Lie algebra over $\mathbb{C}$ with nilradical $\mathfrak{n}$. Fix a vector space decomposition $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$, and choose a basis $\{f_1, \dots, f_d\}$ of $\mathfrak{f}$. For each $j \in \{1, \dots, d\}$, set

$$D_j := \text{ad}(f_j)|_{\mathfrak{n}} \in \mathfrak{gl}(\mathfrak{n}),$$

and write its Jordan–Chevalley decomposition as $D_j = (D_j)_s + (D_j)_n$. Since $\mathfrak{s}$ is solvable, the operators $\{D_j\}$ span a solvable subalgebra of $\mathfrak{gl}(\mathfrak{n})$. In particular, the semisimple parts $\{(D_j)_s\}_{j=1}^d$ commute and are therefore simultaneously diagonalizable over $\mathbb{C}$ (see [Bou05, Theorem 7.1.1]). This proves the following weight space decomposition:

Proposition 2.1. *The nilradical admits a decomposition into common eigenspaces*

$$\mathfrak{n} = \bigoplus_{\alpha \in \Delta} V_{\alpha},$$

where $\Delta \subset \mathfrak{f}^* := \text{Hom}_{\mathbb{C}}(\mathfrak{f}, \mathbb{C})$ is the finite, non-empty set of weights, and

$$V_{\alpha} := \{v \in \mathfrak{n} \mid (D_j)_s(v) = \alpha(f_j)v \text{ for all } j = 1, \dots, d\}.$$

Equivalently, $|\Delta|$ is the number of distinct eigenvalue d -tuples $(\alpha(f_1), \dots, \alpha(f_d))$ arising from the action of $\{(D_j)_s\}$ on $\mathfrak{n}$.

Choose a basis $\mathcal{B} = (x_1, \dots, x_M, f_1, \dots, f_d)$ of $\mathfrak{s}$ adapted to the decomposition $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$, with corresponding indeterminates $(z_{x_1}, \dots, z_{x_M}, z_{f_1}, \dots, z_{f_d})$. For each $\alpha \in \Delta$, define the linear form

$$\ell_{\alpha}(z_{\mathfrak{f}}) := \sum_{j=1}^d \alpha(f_j) z_{f_j}.$$

Recall that the characteristic polynomial is defined as the determinant of the matrix pencil

$$\mathbf{A}(z_0, \mathbf{z}) := z_0 I_N + \sum_{i=1}^N z_i \text{ad}(y_i).$$

With the weight structure established, we use it to study the characteristic polynomial. Our approach is guided by the goal of obtaining a statement analogous to the Dedekind–Frobenius Theorem (Theorem 1.1), where the factorization of the group determinant is indexed by the unitary dual of a finite group. In the Lie algebra setting, it is natural to ask whether the characteristic polynomial $Q_{\mathfrak{s}}(z)$ admits a similar structural interpretation. The theorem below shows that for solvable Lie algebras with nilradical, the weight structure of the nilradical plays an analogous role, as the irreducible linear factors of $Q_{\mathfrak{s}}(z)$ are indexed by the weights arising from the action of the extension, with multiplicities determined by the corresponding weight spaces.

This result also resolves Problem 1.7, posed by [KY21], which asks whether the spectral invariant $k(\mathfrak{s})$ can be expressed in terms of other well-known Lie-theoretic invariants.

Theorem 2.2. *The characteristic polynomial of a solvable Lie algebra with nilradical factors as*

$$Q_{\mathfrak{s}}(z) = z_0^{\dim \mathfrak{f}} \prod_{\alpha \in \Delta} (z_0 + \ell_{\alpha}(z_{\mathfrak{f}}))^{\dim V_{\alpha}}.$$

Furthermore, the spectral invariant is given by

$$k(\mathfrak{s}) = |\Delta \cup \{0\}|.$$

Proof. With respect to the decomposition $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$, the matrix pencil $\mathbf{A}(z_0, \mathbf{z})$ is block upper-triangular:

$$\mathbf{A}(z_0, \mathbf{z}) = \begin{pmatrix} \mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{n}} & * \\ 0 & \mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{f}} \end{pmatrix}.$$

This is because $\mathfrak{n}$ is an ideal, so $[\mathfrak{s}, \mathfrak{n}] \subseteq \mathfrak{n}$, and hence each $\text{ad}(y)$ is block upper-triangular in any basis adapted to the decomposition. Taking determinants gives

$$Q_{\mathfrak{s}}(z) = \det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{n}}) \det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{f}}).$$

We first compute the quotient block $\det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{f}})$, corresponding to the induced action on $\mathfrak{s}/\mathfrak{n} \cong \mathfrak{f}$. First, we claim that $[\mathfrak{s}, \mathfrak{s}] \subseteq \mathfrak{n}$. By Lie's Theorem, there exists a basis of $\mathfrak{s}$ in which all operators $\text{ad}(\mathfrak{s})$ are simultaneously upper-triangular. Commutators of upper-triangular matrices are strictly upper-triangular and hence nilpotent, so $\text{ad}([x, y])$ is nilpotent for all $x, y \in \mathfrak{s}$. By Engel's Theorem, the derived algebra $[\mathfrak{s}, \mathfrak{s}]$ is therefore a nilpotent ideal, and by maximality of the nilradical, $[\mathfrak{s}, \mathfrak{s}] \subseteq \mathfrak{n}$. It follows that the induced adjoint action of $\mathfrak{s}$ on the quotient $\mathfrak{s}/\mathfrak{n}$ is trivial. Hence the corresponding block of the pencil is $z_0 I_{\dim \mathfrak{f}}$, and

$$\det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{f}}) = z_0^{\dim \mathfrak{f}}.$$

This contributes a single linear factor, corresponding to the zero weight.

Next, consider $\det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{n}})$. Choose a basis $\mathcal{B}_{\mathfrak{n}}$ adapted to the weight decomposition $\mathfrak{n} = \bigoplus_{\alpha \in \Delta} V_{\alpha}$, obtained by concatenating bases $\mathcal{B}_{\alpha}$ of each V_{α} and chosen to respect the Jordan–Chevalley decomposition. For $x \in \mathfrak{n}$, Engel's Theorem implies that $\text{ad}(x)|_{\mathfrak{n}}$ is nilpotent, hence strictly upper-triangular. Similarly, for $f_j \in \mathfrak{f}$, the nilpotent part $(D_j)_n$ acts nilpotently on each V_{α} , contributing only strictly upper-triangular terms. Thus, the diagonal entries of $\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{n}}$ arise solely from z_0 and the semisimple parts $(D_j)_s$. For a basis vector $v \in \mathcal{B}_{\alpha}$, the corresponding diagonal entry is

$$z_0 + \sum_{j=1}^d z_{f_j} \alpha(f_j) = z_0 + \ell_{\alpha}(z_f).$$

Taking the product over all basis vectors gives

$$\det(\mathbf{A}(z_0, \mathbf{z})|_{\mathfrak{n}}) = \prod_{\alpha \in \Delta} (z_0 + \ell_{\alpha}(z_f))^{\dim V_{\alpha}}.$$

Combining the two blocks gives the stated factorization:

$$Q_{\mathfrak{s}}(z) = z_0^{\dim \mathfrak{f}} \prod_{\alpha \in \Delta} (z_0 + \ell_{\alpha}(z_f))^{\dim V_{\alpha}}.$$

The distinct linear factors are $\{\ell_{\alpha} \mid \alpha \in \Delta\}$ together with the zero form, so $k(\mathfrak{s}) = |\Delta \cup \{0\}|$. □

Remark 2.3. We can also use this result to recover Theorem 1.5, the characterization of nilpotent Lie algebras studied in [KY21, Mul23]: An n -dimensional Lie algebra L is nilpotent if and only if $Q_L(z)$ is spectrally equivalent to z_0^n . This is because if L is nilpotent, then $L = \mathfrak{n}$ and the weight set is trivial, so the factorization formula gives $Q_L(z) = z_0^n$. Conversely, if $Q_L(z) = z_0^n$, then all weights vanish, which forces $\text{ad}(f)|_{\mathfrak{n}}$ to be nilpotent for every f in a complementary subspace. By maximality of the nilradical, this implies $L = \mathfrak{n}$.

We can use this theorem to obtain sharp bounds for the spectral invariant $k(\mathfrak{s})$.

Theorem 2.4. Let $\mathfrak{s}$ be a solvable Lie algebra with nilradical $\mathfrak{n}$ and extension $\mathfrak{f}$. The spectral invariant $k(\mathfrak{s})$ satisfies the sharp bounds:

$$\dim(\mathfrak{f}) + 1 \leq k(\mathfrak{s}) \leq \dim(\mathfrak{n}) + 1.$$

The equality conditions are:

- Lower Bound: $k(\mathfrak{s}) = \dim(\mathfrak{f}) + 1$ if and only if the set of non-zero weights $\Delta \setminus \{0\}$ forms a basis of $\mathfrak{f}^*$.
- Upper Bound: $k(\mathfrak{s}) = \dim(\mathfrak{n}) + 1$ if and only if $0 \notin \Delta$ and $\dim V_{\alpha} = 1$ for all $\alpha \in \Delta$.

Proof. We prove the two bounds separately:

- Lower Bound: We first show that the set of weights spans the dual space. Suppose $\text{span}(\Delta) \subsetneq \mathfrak{f}^*$. Then there exists a nonzero $f_0 \in \mathfrak{f}$ such that $\alpha(f_0) = 0$ for all $\alpha \in \Delta$. The eigenvalues of $\text{ad}(f_0)|_{\mathfrak{n}}$ are $\{\alpha(f_0) \mid \alpha \in \Delta\} = \{0\}$, so this operator is nilpotent. Moreover, as shown in the proof of Theorem 2.2, the induced action on $\mathfrak{s}/\mathfrak{n}$ is trivial, hence nilpotent. Therefore $\text{ad}(f_0)$ is globally nilpotent on $\mathfrak{s}$, so f_0 belongs to the nilradical $\mathfrak{n}$. This contradicts $f_0 \in \mathfrak{f} \setminus \{0\}$.

Thus $\text{span}(\Delta) = \mathfrak{f}^*$. Since the zero vector does not contribute to the span, the set of non-zero weights $\Delta' := \Delta \setminus \{0\}$ must span $\mathfrak{f}^*$. Consequently, $|\Delta'| \geq \dim(\mathfrak{f})$.

Recall that $k(\mathfrak{s})$ counts the number of distinct linear factors in $Q_{\mathfrak{s}}(z)$. By Theorem 2.2, the extension $\mathfrak{f}$ contributes the factor $z_0^{\dim \mathfrak{f}}$. Thus, the linear form $\ell(z) = 0$ is always a factor of $Q_{\mathfrak{s}}(z)$, regardless of whether $0 \in \Delta$. The set of distinct factors is therefore $\Delta \cup \{0\}$. We can compute the cardinality as:

$$k(\mathfrak{s}) = |\Delta \cup \{0\}| = |\Delta'| + 1.$$

Combining this with the span inequality $|\Delta'| \geq \dim(\mathfrak{f})$, we obtain:

$$k(\mathfrak{s}) \geq \dim(\mathfrak{f}) + 1.$$

Equality holds if and only if $|\Delta'| = \dim(\mathfrak{f})$, which occurs if and only if the non-zero weights Δ' form a basis of $\mathfrak{f}^*$.

- Upper Bound: Since $\mathfrak{n} = \bigoplus_{\alpha \in \Delta} V_{\alpha}$, we have

$$\dim(\mathfrak{n}) = \sum_{\alpha \in \Delta} \dim(V_{\alpha}) \geq \sum_{\alpha \in \Delta} 1 = |\Delta|.$$

If $0 \in \Delta$, then $k(\mathfrak{s}) = |\Delta| \leq \dim(\mathfrak{n}) < \dim(\mathfrak{n}) + 1$. If $0 \notin \Delta$, then $k(\mathfrak{s}) = |\Delta| + 1 \leq \dim(\mathfrak{n}) + 1$. This proves the bound.

Equality requires $0 \notin \Delta$ and $|\Delta| = \dim(\mathfrak{n})$, which holds if and only if $\dim(V_{\alpha}) = 1$ for every $\alpha \in \Delta$. $\square$

The factorization of Theorem 2.2 allows the notion of spectral equivalence to be reformulated as a condition on the collection of weights.

Proposition 2.5. *Let $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$ and $\mathfrak{s}' = \mathfrak{n}' \oplus \mathfrak{f}'$ be solvable Lie algebras. Let $\Delta \subset \mathfrak{f}^*$ and $\Delta' \subset (\mathfrak{f}')^*$ denote the multisets of weights for the adjoint actions of $\mathfrak{f}$ on $\mathfrak{n}$ and $\mathfrak{f}'$ on $\mathfrak{n}'$, respectively. Then $\mathfrak{s} \sim \mathfrak{s}'$ if and only if $\dim(\mathfrak{f}) = \dim(\mathfrak{f}')$ and there exists a linear isomorphism $T: \mathfrak{f}^* \rightarrow (\mathfrak{f}')^*$ such that*

$$\Delta' = \{T(\alpha) \mid \alpha \in \Delta\},$$

where equality holds as multisets.

Proof. We prove the two implications separately.

($\Rightarrow$) Suppose $\mathfrak{s} \sim \mathfrak{s}'$. By definition, there exists an invertible matrix $B \in \text{GL}_N(\mathbb{C})$ such that $Q_{\mathfrak{s}'}(z_0, \mathbf{z}) = Q_{\mathfrak{s}}(z_0, \mathbf{z}B)$. By Theorem 2.2, the linear factors of $Q_{\mathfrak{s}}(z_0, \mathbf{z})$ are of the form $z_0 + \ell_{\alpha}(\mathbf{z})$, where ℓ_{α} corresponds to a weight $\alpha \in \Delta$. Under the substitution $\mathbf{z} \mapsto \mathbf{z}B$, such a factor transforms into $z_0 + \ell_{\alpha}(\mathbf{z}B)$. Let $\ell_{\alpha} \circ B$ denote the linear form $\mathbf{z} \mapsto \ell_{\alpha}(\mathbf{z}B)$. By the uniqueness of polynomial factorization over $\mathbb{C}$, the multiset of linear forms appearing in $Q_{\mathfrak{s}'}(z_0, \mathbf{z})$ must be the multiset $\{\ell_{\alpha} \circ B \mid \alpha \in \Delta \cup \{0\}\}$.

Since the linear forms for $\mathfrak{s}'$ are identified with weights in $(\mathfrak{f}')^*$, the map $\alpha \mapsto \ell_{\alpha} \circ B$ induces a linear map $T: \mathfrak{f}^* \rightarrow (\mathfrak{f}')^*$. Restricted to the extension subspace, this is the dual map induced by the relevant block of B . Since B is invertible and the weights span the dual spaces (or strictly, are defined on the extensions), T defines the required isomorphism such that $\Delta' = T(\Delta)$ as multisets.

($\Leftarrow$) Conversely, suppose there exists a linear isomorphism $T: \mathfrak{f}^* \rightarrow (\mathfrak{f}')^*$ such that $\Delta' = T(\Delta)$ as multisets.

First, T being an isomorphism implies $\dim(\mathfrak{f}) = \dim(\mathfrak{f}')$. Since both algebras have total dimension N , the nilradicals must also satisfy $\dim(\mathfrak{n}) = N - \dim(\mathfrak{f}) = \dim(\mathfrak{n}')$. Let $U: \mathfrak{n}' \rightarrow \mathfrak{n}$ be any vector space isomorphism. Let $S: \mathfrak{f}' \rightarrow \mathfrak{f}$ be the dual isomorphism to T , characterized by $\alpha(Sf') = (T\alpha)(f')$ for all $\alpha \in \mathfrak{f}^*, f' \in \mathfrak{f}'$. We construct the matrix $B \in \text{GL}_N(\mathbb{C})$ to be the block-diagonal operator corresponding

to the map $U \oplus S: \mathfrak{n}' \oplus \mathfrak{f}' \rightarrow \mathfrak{n} \oplus \mathfrak{f}$. We evaluate the characteristic polynomial of $\mathfrak{s}$ at the transformed coordinates $\mathbf{z}B$. For any weight $\alpha \in \Delta$, the linear form ℓ_α transforms as:

$$\ell_\alpha(\mathbf{z}B) = \ell_\alpha((U \oplus S)\mathbf{z}) = \alpha(S\mathbf{z}_{\mathfrak{f}'}) = (T\alpha)(\mathbf{z}_{\mathfrak{f}'}).$$

Therefore, the factors of $Q_{\mathfrak{s}}(z_0, \mathbf{z}B)$ are $\{z_0 + (T\alpha)(\mathbf{z}_{\mathfrak{f}'})\}_{\alpha \in \Delta}$. Since $T(\Delta) = \Delta'$ as multisets, these are the factors of $Q_{\mathfrak{s}'}(z_0, \mathbf{z})$. Thus $Q_{\mathfrak{s}}(z_0, \mathbf{z}B) = Q_{\mathfrak{s}'}(z_0, \mathbf{z})$, and $\mathfrak{s} \sim \mathfrak{s}'$. $\square$

In many classifications of solvable Lie algebras, the families are listed such that they depend on a continuous family of parameters. Let $\mathfrak{s}(\mathbf{p})$ be a family of N -dimensional solvable Lie algebras over $\mathbb{C}$, continuously parameterized by $\mathbf{p} \in \mathcal{P}$, where $\mathcal{P}$ is a connected domain in $\mathbb{C}^{|\mathbf{p}|}$. While all nilpotent Lie algebras of a fixed dimension form a single spectral equivalence class, the case for solvable Lie algebras is much more complex. In this setting, continuous deformations of the structure constants often lead to distinct spectral equivalence classes, even when discrete invariants such as $k(\mathfrak{s})$ remain constant. This phenomenon shows that discrete invariants may not give a sufficiently fine classification and necessitates other approaches to distinguish members of continuous families. We introduce the concept of spectral rigidity to identify cases where given a family of solvable Lie algebras, the characteristic polynomial uniquely recovers the defining parameters of each member.

Definition 2.6. A parameterized family of Lie algebras $\mathfrak{s}(\mathbf{p})$ is **spectrally rigid** if for any $\mathbf{p}, \mathbf{p}' \in \mathcal{P}$, the condition $\mathfrak{s}(\mathbf{p}) \sim \mathfrak{s}(\mathbf{p}')$ implies $\mathbf{p} = \mathbf{p}'$.

By Proposition 2.5, $\mathfrak{s}(\mathbf{p}) \sim \mathfrak{s}(\mathbf{p}')$ holds if and only if $\Delta(\mathbf{p})$ and $\Delta(\mathbf{p}')$ lie in the same $\text{GL}(\mathfrak{f}^*)$ -orbit. This proves:

Corollary 2.7. Let $\mathfrak{s}(\mathbf{p})$ be a family of solvable Lie algebras parameterized by $\mathbf{p} \in \mathcal{P}$, and let $\Delta(\mathbf{p}) \subset \mathfrak{f}^*$ denote the corresponding multiset of weights. The family $\mathfrak{s}(\mathbf{p})$ is rigid if and only if for any $\mathbf{p} \neq \mathbf{p}'$ there does not exist a linear automorphism $T \in \text{GL}(\mathfrak{f}^*)$ such that

$$\Delta(\mathbf{p}') = \{T(\alpha) \mid \alpha \in \Delta(\mathbf{p})\}.$$

2.2 Topological Spectral Invariants

We now study the topological consequences of the factorization in Theorem 2.2. Recall that the eigenvariety $V_{\mathfrak{s}} \subset \mathbb{C}^{N+1}$ is the zero locus of the characteristic polynomial $Q_{\mathfrak{s}}(z_0, \mathbf{z})$. As discussed in the introduction, $V_{\mathfrak{s}}$ always contains the hyperplane $\{z_0 = 0\}$, which we want to exclude. Therefore, we restrict our attention to the variety $V_{\mathfrak{s}}^*$, defined as the zero locus of the polynomial factor $p(z_0, \mathbf{z})$ where $Q_{\mathfrak{s}}(z_0, \mathbf{z}) = z_0^k p(z_0, \mathbf{z})$. Our goal is to compute the Poincaré polynomial $P_{\mathfrak{s}}(t)$ of $(V_{\mathfrak{s}}^*)^c \subset \mathbb{C}^N$.

Let Φ be the set of distinct linear factors of $p(z_0, \mathbf{z})$. We define the hyperplane arrangement $\mathcal{A}_{\mathfrak{s}}$ as the collection of hyperplanes $\{\ker L \mid L \in \Phi\}$ in $\mathbb{C}^{N+1}$. The complement $M(\mathcal{A}_{\mathfrak{s}}) = \mathbb{C}^{N+1} \setminus \bigcup_{H \in \mathcal{A}_{\mathfrak{s}}} H$ is the relevant manifold.

For each linear form $\ell \in \Phi$, we define the associated logarithmic 1-form:

$$\omega_{\ell} = \frac{1}{2\pi i} \frac{d\ell}{\ell}.$$

Let $\mathfrak{A}$ be the graded $\mathbb{C}$ -subalgebra of forms on $M(\mathcal{A}_{\mathfrak{s}})$ generated by $\{1\} \cup \{\omega_{\ell}\}_{\ell \in \Phi}$, and $[w_{\ell}]$ be the de Rham cohomology class associated with w_{ℓ} .

Orlik and Solomon [OS80] determined the structure of $\mathfrak{A}$ entirely from the combinatorial dependencies among the linear forms. We first recall their general result.

Theorem 2.8 ([OS80], Theorem 5.2). *Let $\mathcal{A}$ be a finite set of hyperplanes in $\mathbb{C}^{\ell}$. There exists an isomorphism $\mathfrak{A} \simeq H^*(M)$ of graded algebras such that $a_A \rightarrow [\omega_A]$ for all $A \in \mathcal{A}$.*

To apply this, we first explicitly construct the algebra $\mathfrak{A}$ for our hyperplane arrangement using the presentation defined in [OS80]. The algebra $\mathfrak{A}$ in Theorem 2.8 is defined in [OS80] as a quotient E/I . By identifying the ambient space $\mathbb{C}^{N+1}$ with $\mathbb{C}^{\ell}$ and the hyperplane arrangement $\mathcal{A}_{\mathfrak{s}}$ with $\mathcal{A}$, we obtain the following:

Corollary 2.9. *Let $\mathcal{A}_{\mathfrak{s}}$ be the hyperplane arrangement defined by the factors of $p(z_0, \mathbf{z})$. Let E be the exterior algebra generated by $\{e_{\ell}\}_{\ell \in \Phi}$. The cohomology ring $H^*(M(\mathcal{A}_{\mathfrak{s}}), \mathbb{C})$ is isomorphic to the quotient*

E/I , where I is the ideal generated by $\partial(e_S)$ for all dependent subsets $S \subseteq \Phi$. For any dependent subset $S = \{\ell_1, \dots, \ell_k\} \subseteq \Phi$ (with $\ell_1 < \dots < \ell_k$ in the fixed ordering), the generator is given by:

$$\partial(e_S) := \sum_{j=1}^k (-1)^{j-1} e_{\ell_1} \wedge \dots \wedge \widehat{e_{\ell_j}} \wedge \dots \wedge e_{\ell_k}$$

where $\widehat{e_{\ell_j}}$ indicates the omission of that term.

This corollary reduces the calculation of the cohomology ring to the combinatorics of the linear dependencies. We can study this combinatorics using matroids.

Definition 2.10 (Spectral Matroid). *The **spectral matroid** $\mathcal{M}_{\mathfrak{s}} = (\Phi, \mathcal{I})$ is the linear matroid on the ground set Φ , where a subset $S \subseteq \Phi$ is independent ($S \in \mathcal{I}$) if and only if the linear forms in S are linearly independent in the dual space $(\mathbb{C} \oplus \mathfrak{f})^*$.*

A nice property of the spectral matroid is that it is an invariant of the spectral equivalence class.

Lemma 2.11. *The spectral matroid $\mathcal{M}_{\mathfrak{s}}$ is a well-defined linear matroid. Furthermore, if $\mathfrak{s} \sim \mathfrak{s}'$, then their spectral matroids are isomorphic: $\mathcal{M}_{\mathfrak{s}} \cong \mathcal{M}_{\mathfrak{s}'}$.*

Proof. The first assertion follows directly from standard linear algebra. For the second assertion, suppose that $\mathfrak{s} \sim \mathfrak{s}'$. By Proposition 2.5, there exists a linear isomorphism $T: \mathfrak{f}^* \rightarrow (\mathfrak{f}')^*$ mapping the multiset of weights Δ bijectively onto Δ' . Extend T to a linear isomorphism $\Psi: (\mathbb{C} \oplus \mathfrak{f})^* \rightarrow (\mathbb{C} \oplus \mathfrak{f}')^*$ by requiring $\Psi(z_0) = z_0$ and $\Psi(\alpha) = T(\alpha)$ for all $\alpha \in \mathfrak{f}^*$. The ground set Φ of $\mathcal{M}_{\mathfrak{s}}$ consists of the linear forms $\ell_0 = z_0$ and $\ell_{\alpha} = z_0 + \alpha$. The action of Ψ on these elements is given by $\Psi(\ell_0) = z_0 = \ell'_0$ and $\Psi(\ell_{\alpha}) = z_0 + T(\alpha) = \ell'_{T(\alpha)}$. Since T induces a bijection $\Delta \rightarrow \Delta'$, the map Ψ induces a bijection between the ground sets Φ and Φ' . Because Ψ is a linear isomorphism, it preserves all linear dependence relations. Thus, Ψ induces an isomorphism of matroids $\mathcal{M}_{\mathfrak{s}} \cong \mathcal{M}_{\mathfrak{s}'}$. $\square$

This allows us to connect everything back to the cohomology neatly:

Proposition 2.12. *For any finite-dimensional solvable Lie algebra $\mathfrak{s}$, the cohomology ring $H^*((V_{\mathfrak{s}}^*)^c, \mathbb{C})$ is a spectral invariant determined uniquely by the spectral matroid $\mathcal{M}_{\mathfrak{s}}$.*

Proof. Let $\mathfrak{s} \sim \mathfrak{s}'$, so there exists a linear isomorphism $T: \mathfrak{f}^* \rightarrow (\mathfrak{f}')^*$ mapping the nilradical weight multiset Δ bijectively onto Δ' . This correspondence induces a bijection between the sets of linear factors Φ and Φ' , thereby identifying the associated hyperplane arrangements $\mathcal{A}_{\mathfrak{s}}$ and $\mathcal{A}_{\mathfrak{s}'}$.

By Lemma 2.11, the linear dependence relations in Φ are given by the spectral matroid $\mathcal{M}_{\mathfrak{s}}$, and the isomorphism T preserves these relations, giving $\mathcal{M}_{\mathfrak{s}} \cong \mathcal{M}_{\mathfrak{s}'}$. Consequently, the collections of dependent subsets defining the Orlik–Solomon ideal I correspond under this bijection.

By Corollary 2.9, the singular cohomology algebra $H^*(M(\mathcal{A}_{\mathfrak{s}}), \mathbb{C})$ is isomorphic to the Orlik–Solomon algebra $A(\mathcal{M}_{\mathfrak{s}}) = E/I$. Since $A(\mathcal{M}_{\mathfrak{s}}) \cong A(\mathcal{M}_{\mathfrak{s}'})$, the cohomology of the complement $M(\mathcal{A}_{\mathfrak{s}})$ is a spectral invariant. As $(V_{\mathfrak{s}}^*)^c$ is the base of a trivial bundle over $M(\mathcal{A}_{\mathfrak{s}})$ with fixed fiber $\mathbb{C}^*$, its cohomology ring is determined by the Orlik–Solomon algebra, so it is a well-defined spectral invariant. $\square$

To compute the Betti numbers of $(V_{\mathfrak{s}}^*)^c$ explicitly, we introduce the some tools related to the intersection lattice.

Definition 2.13 (Intersection Lattice). *Let $\mathcal{A}_{\mathfrak{s}}$ be the hyperplane arrangement defined by the linear factors of $p(z_0, \mathbf{z})$. The **intersection lattice** $L(\mathcal{A}_{\mathfrak{s}})$ consists of all subspaces $X \subseteq \mathbb{C}^{N+1}$ of the form $X = \bigcap_{H \in S} H$ for some $S \subseteq \mathcal{A}_{\mathfrak{s}}$. The partial order is given by reverse inclusion: $X \leq Y$ if $Y \subseteq X$. The minimal element is $\hat{0} = \mathbb{C}^{N+1}$, and the maximal element is $\hat{1} = \bigcap_{H \in \mathcal{A}_{\mathfrak{s}}} H$. The rank function is defined by $\text{rank}(X) = \text{codim}_{\mathbb{C}}(X)$.*

The inclusion–exclusion structure of the arrangement is given by the Möbius function of this lattice.

Definition 2.14 (Möbius Function). *The **Möbius function** $\mu: L(\mathcal{A}_{\mathfrak{s}}) \times L(\mathcal{A}_{\mathfrak{s}}) \rightarrow \mathbb{Z}$ is characterized by $\mu(X, X) = 1$ for all X , and by the relation*

$$\sum_{X \leq Z \leq Y} \mu(X, Z) = 0 \quad \text{for } X < Y.$$

We write $\mu(X) = \mu(\hat{0}, X)$.

To count things exactly, we also need to introduce Whitney numbers.

Definition 2.15 (Whitney Numbers). *The **Whitney numbers of the first kind** are defined by*

$$w_k = \sum_{\text{rank}(X)=k} |\mu(X)|.$$

Additionally, recall that an arrangement $\mathcal{A}$ in a vector space V is called central if the intersection of all hyperplanes in $\mathcal{A}$ contains the origin: $\bigcap_{H \in \mathcal{A}} H \neq \emptyset$. It is straightforward to show that the hyperplane arrangement $\mathcal{A}_s$ is central.

To compute the Betti numbers explicitly, we rely on the combinatorial formula established by Orlik and Solomon. To state their general result relating the algebra dimensions to the Möbius function, we need to recall some facts about finite geometric lattices. Recall that a finite lattice L is a geometric lattice if it is both atomistic and semimodular. A lattice is atomistic if every element $X \in L$ is the join of atoms, where an atom is an element $A \in L$ such that $\hat{0} < A$ and there is no element Y satisfying $\hat{0} < Y < A$. In the intersection lattice $L(\mathcal{A}_s)$, the atoms correspond to the hyperplanes $H \in \mathcal{A}_s$. The lattice is semimodular if for any two elements $X, Y \in L$, the rank function satisfies the inequality $\text{rank}(X \vee Y) + \text{rank}(X \wedge Y) \leq \text{rank}(X) + \text{rank}(Y)$, where the join $X \vee Y$ corresponds to the intersection $X \cap Y$ and the meet $X \wedge Y$ corresponds to the subspace $\text{span } X \cup Y$. This structure ensures that the rank function is well-defined as the length of the maximal chain from $\hat{0}$ to X , which in our case is identically the codimension of the subspace X in $\mathbb{C}^{N+1}$.

Theorem 2.16 ([OS80], Theorem 2.6). *Let L be a finite geometric lattice. The Poincaré polynomial of the associated graded algebra $\mathfrak{A}$ is given by*

$$P_{\mathfrak{A}}(t) = \sum_{X \in L} \mu(X) (-t)^{\text{rank}(X)},$$

where $\mu(X) = \mu(\hat{0}, X)$ denotes the Möbius function of the lattice.

We apply this result to the intersection lattice $L(\mathcal{A}_s)$. Since the hyperplane arrangement is central, its intersection lattice is a finite geometric lattice, and the rank corresponds to the codimension of the subspaces.

Proposition 2.17. *The k -th Betti number of the complement $M(\mathcal{A}_s)$ is given by the k -th Whitney number of the spectral matroid:*

$$b_k(M) = w_k := \sum_{\text{rank}(X)=k} |\mu(X)|.$$

Proof. As established in Corollary 2.9, we have an isomorphism of graded algebras $H^*(M(\mathcal{A}_s), \mathbb{C}) \cong \mathfrak{A}$. Consequently, the Poincaré polynomial $\pi(t) = \sum b_k(M) t^k$ coincides with $P_{\mathfrak{A}}(t)$ from Theorem 2.16. For geometric lattices, the Möbius function satisfies the sign property $\text{sgn}(\mu(X)) = (-1)^{\text{rank}(X)}$, which allows the term $\mu(X) (-t)^{\text{rank}(X)}$ to be rewritten as $|\mu(X)| t^{\text{rank}(X)}$. Summing over all elements of rank k gives the k -th Whitney number w_k as the coefficient of t^k . $\square$

Having determined the cohomology of the vector space complement $M(\mathcal{A}_s)$ via the Orlik–Solomon algebra, we now turn to the topological invariants. To isolate the non-trivial topology, we restrict our attention to the hyperplane $z_0 = 1$. In this section, we identify $(V_s^*)^c$ with the complement of the eigenvariety restricted to this slice (geometrically, the complement of the deconing $d\mathcal{A}_s$). Let $\mathcal{A}$ be a central arrangement. The deconing of $\mathcal{A}$ with respect to the coordinate z_0 is the arrangement $d\mathcal{A}$ obtained by intersecting the hyperplanes of $\mathcal{A}$ with the hyperplane $H = \{v \in V \mid z_0(v) = 1\}$.

Proposition 2.18 ([OT92], Proposition 5.1). *Let $\mathcal{A}$ be a central arrangement. The restriction of the Hopf bundle to the complement $M(\mathcal{A})$ is trivial, giving a diffeomorphism*

$$M(\mathcal{A}) \cong M(d\mathcal{A}) \times \mathbb{C}^*,$$

where $M(d\mathcal{A})$ denotes the complement of the deconing of $\mathcal{A}$.

Corollary 2.19. *Let $\mathcal{A}_{\mathfrak{s}}$ be the central hyperplane arrangement defined by the homogeneous polynomial $p(z_0, \mathbf{z})$. Identifying $(V_{\mathfrak{s}}^*)^c$ with the affine complement $M(d\mathcal{A}_{\mathfrak{s}})$, we have the diffeomorphism:*

$$M(\mathcal{A}_{\mathfrak{s}}) \cong (V_{\mathfrak{s}}^*)^c \times \mathbb{C}^*.$$

Proof. The arrangement $\mathcal{A}_{\mathfrak{s}}$ is central. By Proposition 2.18, we have the decomposition $M(\mathcal{A}_{\mathfrak{s}}) \cong M(d\mathcal{A}_{\mathfrak{s}}) \times \mathbb{C}^*$. Since we identified $(V_{\mathfrak{s}}^*)^c$ with the fiber complement $M(d\mathcal{A}_{\mathfrak{s}})$ in the affine chart $z_0 = 1$, the result follows immediately. $\square$

This gives us the following result:

Theorem 2.20. *Let $P_{\mathfrak{s}}(t)$ denote the Poincaré polynomial of the complement $(V_{\mathfrak{s}}^*)^c$. Then $P_{\mathfrak{s}}(t)$ is determined by the Whitney numbers of the spectral matroid $\mathcal{M}_{\mathfrak{s}}$ via the formula:*

$$P_{\mathfrak{s}}(t) = \frac{\sum_{k \geq 0} w_k t^k}{1+t}.$$

Proof. By Theorem 2.19, there is a diffeomorphism $M(\mathcal{A}_{\mathfrak{s}}) \cong (V_{\mathfrak{s}}^*)^c \times \mathbb{C}^*$. We apply the Künneth formula for the cohomology of a product space. Let $\pi(t)$ denote the Poincaré polynomial of the vector space complement $M(\mathcal{A}_{\mathfrak{s}})$. We have:

$$\pi(t) = P_{\mathfrak{s}}(t) \cdot P_{\mathbb{C}^*}(t).$$

The cohomology of the torus $\mathbb{C}^*$ is generated by the identity and a single degree-1 class, so $P_{\mathbb{C}^*}(t) = 1+t$. Furthermore, by Proposition 2.17, the polynomial $\pi(t)$ is given by the generating function of the Whitney numbers $\sum w_k t^k$. Substituting these into the Künneth relation gives:

$$\sum_{k \geq 0} w_k t^k = P_{\mathfrak{s}}(t) \cdot (1+t).$$

Solving for $P_{\mathfrak{s}}(t)$ completes the proof. $\square$

This factorization allows us to derive an explicit combinatorial formula for the spectral Betti numbers $b_j(\mathfrak{s})$. This proves that the Betti numbers $b_j(\mathfrak{s})$ are determined by the weights of the nilradical, which answers Problem 1.13.

Theorem 2.21. *The j -th Betti number is given by the alternating sum of the Whitney numbers up to rank j of the spectral matroid:*

$$b_j(\mathfrak{s}) = \sum_{i=0}^j (-1)^{j-i} w_i.$$

Proof. We determine the coefficients $b_j(\mathfrak{s})$ by expanding the polynomial relation from Theorem 2.20:

$$\sum_{k \geq 0} w_k t^k = (1+t) \sum_{j \geq 0} b_j(\mathfrak{s}) t^j = \sum_{j \geq 0} b_j(\mathfrak{s}) t^j + \sum_{j \geq 0} b_j(\mathfrak{s}) t^{j+1}.$$

Comparing the coefficient of t^k on both sides gives the recurrence relation:

$$w_k = b_k(\mathfrak{s}) + b_{k-1}(\mathfrak{s}),$$

with the boundary condition $b_{-1}(\mathfrak{s}) = 0$. Since the minimal element of the intersection lattice is unique, $w_0 = 1$, which implies $b_0(\mathfrak{s}) = 1$. Rearranging the recurrence as $b_k(\mathfrak{s}) = w_k - b_{k-1}(\mathfrak{s})$, we prove the closed-form expression by induction.

For $j = 0$, the formula gives $(-1)^0 w_0 = 1$, which holds. Assume the formula holds for $j = m-1$. Then:

$$\begin{aligned} b_m(\mathfrak{s}) &= w_m - b_{m-1}(\mathfrak{s}) \\ &= w_m - \sum_{i=0}^{m-1} (-1)^{(m-1)-i} w_i \\ &= w_m + \sum_{i=0}^{m-1} (-1)^{m-i} w_i \\ &= \sum_{i=0}^m (-1)^{m-i} w_i. \end{aligned}$$

This proves the formula for all $j \geq 0$. $\square$

We may now determine the exact degree of the topological spectral invariant $P_{\mathfrak{s}}(t)$ through the nilradical's weight decomposition. This improves the best known result (Proposition 1.12) from an upper bound to an equality.

Theorem 2.22. *Let $\Delta \subset \mathfrak{f}^*$ be the multiset of nilradical weights. The degree of the Poincaré polynomial of the complement $(V_{\mathfrak{s}}^*)^c$ is equal to the dimension of the subspace spanned by the weights:*

$$\deg P_{\mathfrak{s}}(t) = \dim(\text{span}(\Delta)).$$

Proof. According to Theorem 2.16, the degree of the polynomial $\pi(t)$ (the Poincaré polynomial of $M(\mathcal{A}_{\mathfrak{s}})$) is equal to the rank of the maximal element in the intersection lattice $L(\mathcal{A}_{\mathfrak{s}})$. In a linear matroid, this rank corresponds to the dimension of the subspace spanned by the defining forms Φ .

Recall that the set of forms is $\Phi = \{\ell_0\} \cup \{\ell_0 + \ell_{\alpha}\}_{\alpha \in \Delta} \subset (\mathbb{C} \oplus \mathfrak{f})^*$, where $\ell_0 = z_0$ corresponds to the spectral parameter and ℓ_{α} are the weights in $\mathfrak{f}^*$. We decompose the dual space as $(\mathbb{C} \oplus \mathfrak{f})^* \cong \mathbb{C}\ell_0 \oplus \mathfrak{f}^*$. The forms in Φ span the same subspace as the set $\{\ell_0\} \cup \{\ell_{\alpha}\}_{\alpha \in \Delta}$. Since ℓ_0 is linearly independent of the subspace $\mathfrak{f}^*$ containing the weights, the dimension of the total span is:

$$\dim(\text{span}(\Phi)) = \dim(\mathbb{C}\ell_0) + \dim(\text{span}(\Delta)) = 1 + \dim(\text{span}(\Delta)).$$

Thus, $\deg \pi(t) = 1 + \dim(\text{span}(\Delta))$. From the factorization $P_{\mathfrak{s}}(t) = \pi(t)/(1+t)$ established in Theorem 2.20, the degree decreases by 1. The conclusion follows. $\square$

We now derive explicit bounds for the Betti numbers $b_j(\mathfrak{s})$. We utilize the following classical result regarding the topology of arrangements in general position.

Lemma 2.23 ([OS80], Lemma 3.1). *Let $\mathcal{A}$ be an arrangement of m hyperplanes. The Betti numbers $b_j(\mathcal{A})$ are bounded above by the Betti numbers of an arrangement of m hyperplanes in general position. Specifically:*

$$b_j(\mathcal{A}) \leq \binom{m}{j}.$$

We can use this to generalize Proposition 1.11.

Corollary 2.24. *Let $k(\mathfrak{s})$ be the spectral invariant counting the number of distinct linear factors in $Q_{\mathfrak{s}}(z_0, \mathbf{z})$. For all $j \geq 0$, the Betti numbers satisfy:*

$$b_j(\mathfrak{s}) \leq \binom{k(\mathfrak{s}) - 1}{j}.$$

Proof. Let $m = k(\mathfrak{s})$. The arrangement $\mathcal{A}_{\mathfrak{s}}$ consists of m hyperplanes in $\mathbb{C}^{N+1}$. As shown in Theorem 2.20, the Poincaré polynomial of $(V_{\mathfrak{s}}^*)^c$ is related to that of $M(\mathcal{A}_{\mathfrak{s}})$ by $P_{\mathfrak{s}}(t) = P_M(t)/(1+t)$.

Consider an arrangement $\mathcal{A}_{\text{gen}}$ of m hyperplanes in general position. Its Poincaré polynomial is $P_{\text{gen}}(t) = (1+t)^m$. The polynomial for the corresponding quotient space would be $(1+t)^{m-1} = \sum \binom{m-1}{j} t^j$. Since the Betti numbers of the arrangement complement $M(\mathcal{A}_{\mathfrak{s}})$ are bounded by those of $M(\mathcal{A}_{\text{gen}})$ (Lemma 2.23), the Betti numbers of the quotient space $(V_{\mathfrak{s}}^*)^c$ are similarly bounded by the coefficients of the polynomial $(1+t)^{m-1}$. Thus, $b_j(\mathfrak{s}) \leq \binom{m-1}{j}$. $\square$

Recall that a sequence of non-negative real numbers $a_0, a_1, \dots, a_n$ is said to be log-concave if it satisfies the inequality $a_k^2 \geq a_{k-1}a_{k+1}$ for all $1 \leq k < n$. If the sequence has no internal zeros, log-concavity implies unimodality, meaning the sequence increases to a maximum and then decreases. Finally, we prove the log-concavity of the Betti numbers. To apply the result of Huh and Katz [HK12], we first recall the necessary matroid-theoretic definitions.

Definition 2.25 (Characteristic Polynomial of a Matroid). *Let $\mathcal{M}$ be a matroid of rank r . The **characteristic polynomial** of $\mathcal{M}$ is defined via the Möbius function of its lattice as:*

$$\chi_{\mathcal{M}}(q) = \sum_{X \in L(\mathcal{M})} \mu(\hat{0}, X) q^{r - \text{rank}(X)}.$$

Definition 2.26 (Normalized Characteristic Polynomial). *The **normalized characteristic polynomial** is defined as $\bar{\chi}_{\mathcal{M}}(q) = \chi_{\mathcal{M}}(q)/(q-1)$.*

We use the following result from algebraic geometry concerning these polynomials.

Lemma 2.27 ([HK12], Theorem 1.1). *Let $\mathcal{M}$ be a matroid representable over a field of characteristic 0. The coefficients of its normalized characteristic polynomial $\bar{\chi}_{\mathcal{M}}(q)$ form a log-concave sequence.*

We now connect our topological invariant to this algebraic object.

Lemma 2.28. *The Poincaré polynomial $P_{\mathfrak{s}}(t)$ of the complement $(V_{\mathfrak{s}}^*)^c$ and the normalized characteristic polynomial $\bar{\chi}_{\mathcal{M}_{\mathfrak{s}}}(q)$ of the spectral matroid are related by the transformation:*

$$\bar{\chi}_{\mathcal{M}_{\mathfrak{s}}}(-t^{-1}) = (-1)^{r-1} t^{-(r-1)} P_{\mathfrak{s}}(t),$$

where r is the rank of $\mathcal{M}_{\mathfrak{s}}$. Consequently, the sequence of Betti numbers $(b_0, b_1, \dots)$ corresponds to the absolute values of the coefficients of $\bar{\chi}_{\mathcal{M}_{\mathfrak{s}}}(q)$.

Proof. Using the definition of the Whitney numbers $w_k = \sum_{\text{rank}(X)=k} |\mu(\hat{0}, X)|$, the characteristic polynomial can be written as:

$$\chi_{\mathcal{M}_{\mathfrak{s}}}(q) = \sum_{k=0}^r (-1)^k w_k q^{r-k} = q^r \sum_{k=0}^r w_k (-q^{-1})^k.$$

Recall from Theorem 2.16 that the Poincaré polynomial of $M(\mathcal{A}_{\mathfrak{s}})$ is $\pi(t) = \sum w_k t^k$. This gives the identity $\chi_{\mathcal{M}_{\mathfrak{s}}}(q) = (-1)^r q^r \pi(-q^{-1})$.

Substituting $q = -t^{-1}$ into the definition of the normalized polynomial $\bar{\chi}_{\mathcal{M}_{\mathfrak{s}}}(q) = \chi_{\mathcal{M}_{\mathfrak{s}}}(q)/(q-1)$:

$$\bar{\chi}_{\mathcal{M}_{\mathfrak{s}}}(-t^{-1}) = \frac{(-t)^{-r} \pi(t)}{-(1+t)/t} = (-1)^{r-1} t^{-(r-1)} \frac{\pi(t)}{1+t}.$$

By Theorem 2.20, $P_{\mathfrak{s}}(t) = \pi(t)/(1+t)$. The relation follows immediately. $\square$

Theorem 2.29. *The sequence of Betti numbers $b_0(\mathfrak{s}), b_1(\mathfrak{s}), \dots$ is log-concave:*

$$b_{j-1}(\mathfrak{s}) b_{j+1}(\mathfrak{s}) \leq b_j(\mathfrak{s})^2 \quad \text{for all } j \geq 1.$$

Proof. The spectral matroid $\mathcal{M}_{\mathfrak{s}}$ is defined by a set of linear forms in a complex vector space, which implies it is representable over a field of characteristic 0. By Lemma 2.27, the coefficients of the normalized characteristic polynomial form a log-concave sequence. By Lemma 2.28, these coefficients are the Betti numbers $b_j(\mathfrak{s})$. Therefore, the topological spectral invariants are log-concave. $\square$

3 The Heisenberg Case

3.1 The Spectral Invariant $k(L)$

Now, we specialize our theory to study solvable Lie algebras $\mathfrak{s}$ such that $\text{Nil}(\mathfrak{s}) \cong \mathfrak{h}(m)$, which is a $2m+1$ dimensional nilpotent Lie algebra. If $\mathfrak{s}$ has dimension $2m+n+1$, we refer to it as a solvable Lie algebra with nilradical $\mathfrak{h}(m)$ and extension dimension n . The low-dimensional cases, corresponding to $\mathfrak{h}(1)$ and $\mathfrak{h}(2)$, have previously been classified into families, some discrete and others depending on continuous parameters. The findings of [RW93] can be summarized as follows:

$(\dim \mathfrak{h}, n)$	Total Dimension	Number of Families	1-Param Families	2-Param Families
$(3, 1)$	4	3	1	0
$(3, 2)$	5	1	0	0
$(5, 1)$	6	8	3	1
$(5, 2)$	7	8	2	1
$(5, 3)$	8	1	0	0

We denote the t -th family of solvable extensions with Heisenberg nilradical $\mathfrak{h}(m)$ and extension dimension n , depending on s parameters, as $\mathfrak{s}_{2m+1,n}^{s,t}$.

Our study relies on the structure theorem of [RW93], which provides a canonical normal form for all solvable extensions of the Heisenberg algebra.

Theorem 3.1 ([RW93]). *Every indecomposable solvable Lie algebra $\mathfrak{s}$ over $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$ whose nilradical is the Heisenberg algebra $\mathfrak{h}(m)$ admits a canonical basis*

$$(\mathfrak{h}, p_1, \dots, p_m, q_1, \dots, q_m, f_1, \dots, f_f),$$

with commutation relations given, in addition to those of $\mathfrak{h}(m)$, by

$$([f_\alpha, h], [f_\alpha, \xi]) = (h, \xi)M_\alpha, \quad [f_\alpha, f_\beta] = r_{\alpha\beta}h,$$

where $\xi = (p_1, \dots, p_m, q_1, \dots, q_m)$ and

$$M_\alpha = \begin{pmatrix} 2a_\alpha & 0 \\ 0 & a_\alpha I_{2m} + X_\alpha \end{pmatrix}.$$

The constants satisfy $a_1 \in \{0, 1\}$ and $a_2 = \dots = a_f = 0$, while the matrices $X_1, \dots, X_f \in \mathfrak{sp}(2m, \mathbb{F})$ commute pairwise, $[X_\alpha, X_\beta] = 0$. If $a_1 = 0$ (resp. $a_1 = 1$), then $\{X_1, \dots, X_f\}$ (resp. $\{X_2, \dots, X_f\}$) is linearly nilindependent. Furthermore, the constants $r_{\alpha\beta}$ satisfy $r_{\alpha\beta} = 0$ for all α, β when $a_1 = 1$, and $r_{\alpha\beta} = -r_{\beta\alpha} \in \mathbb{F}$ for all α, β when $a_1 = 0$.

This normal form allows us to separate the contribution of the extension from the nontrivial symplectic action on the nilradical.

Lemma 3.2. *Let $\mathfrak{s}$ be a solvable Lie algebra with Heisenberg nilradical $\mathfrak{h}(m)$ and k -dimensional extension $\mathfrak{f}$. Then the characteristic polynomial factors as*

$$Q_{\mathfrak{s}}(z) = z_0^k \left(z_0 + \sum_{\alpha=1}^k 2a_\alpha z_{f_\alpha} \right) \det \left(z_0 I_{2m} + \sum_{\alpha=1}^k z_{f_\alpha} (a_\alpha I_{2m} + X_\alpha) \right).$$

Proof. We fix a basis $\mathcal{B}$ for $\mathfrak{s}$ adapted to the direct sum decomposition $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$, where the nilradical is $\mathfrak{n} = \mathfrak{h}(m) = \text{span}\{h, p_i, q_i\}$ and the extension is $\mathfrak{f} = \text{span}\{f_1, \dots, f_k\}$. The characteristic polynomial is given by $Q_{\mathfrak{s}}(z) = \det(\mathcal{A}(z))$, where the matrix pencil is defined as:

$$\mathcal{A}(z) = z_0 I + \sum_{y \in \mathcal{B}} z_y \text{ad}(y).$$

Since the nilradical $\mathfrak{n}$ is an ideal, the adjoint matrices are block upper-triangular with respect to the decomposition $\mathfrak{s} = \mathfrak{n} \oplus \mathfrak{f}$. Consequently, the pencil takes the form:

$$\mathcal{A}(z) = \begin{pmatrix} \mathcal{A}(z)|_{\mathfrak{n}} & * \\ 0 & \mathcal{A}(z)|_{\mathfrak{f}} \end{pmatrix}.$$

The determinant is the product of the determinants of these diagonal blocks, $Q_{\mathfrak{s}}(z) = \det(\mathcal{A}(z)|_{\mathfrak{n}}) \cdot \det(\mathcal{A}(z)|_{\mathfrak{f}})$. We analyze each contribution separately:

1. Contribution from the $\mathfrak{f}$ -block: The Lie bracket $[f_\alpha, f_\beta] = r_{\alpha\beta}h$ lies strictly within the nilradical $\mathfrak{n}$. Therefore, the induced action of any extension element f_α on the quotient space $\mathfrak{s}/\mathfrak{n} \cong \mathfrak{f}$ is zero. Furthermore, the action of any element from the nilradical on this quotient is trivially zero. Thus, the matrix pencil restricted to the extension block simplifies to the scalar matrix $\mathcal{A}(z)|_{\mathfrak{f}} = z_0 I_k$. Its determinant is:

$$\det(\mathcal{A}(z)|_{\mathfrak{f}}) = \det(z_0 I_k) = z_0^k.$$

2. Contribution from the $\mathfrak{n}$ -block: Restricting to the nilradical, the pencil is $\mathcal{A}(z)|_{\mathfrak{n}} = z_0 I_{2m+1} + \sum z_y \text{ad}(y)|_{\mathfrak{n}}$. By Engel's theorem, for any element $x \in \mathfrak{n}$, the operator $\text{ad}(x)|_{\mathfrak{n}}$ is nilpotent. In a suitable basis, its matrix representation is strictly upper-triangular and contributes zero to the diagonal trace elements. The diagonal entries of the pencil are therefore determined exclusively by z_0 and the semisimple parts of the extension operators $\text{ad}(f_\alpha)|_{\mathfrak{n}}$.

Applying Theorem 3.1, the action of $\text{ad}(f_\alpha)$ on $\mathfrak{n} = \text{span}\{h\} \oplus \text{span}\{p_i, q_i\}$ decomposes into a block diagonal form. The matrix $\text{ad}(f_\alpha)|_{\mathfrak{n}}$ is similar to $\text{diag}(2a_\alpha, a_\alpha I_{2m} + X_\alpha)$. Summing over the parameters, the matrix pencil on the nilradical is similar to:

$$\mathcal{A}(z)|_{\mathfrak{n}} \cong \begin{pmatrix} z_0 + \sum 2a_\alpha z_{f_\alpha} & 0 \\ 0 & z_0 I_{2m} + \sum z_{f_\alpha} (a_\alpha I_{2m} + X_\alpha) \end{pmatrix}.$$

Computing the determinant of this block-diagonal matrix gives:

$$\det(\mathcal{A}(z)|_{\mathfrak{n}}) = \left(z_0 + \sum_{\alpha=1}^k 2a_\alpha z_{f_\alpha} \right) \cdot \det \left(z_0 I_{2m} + \sum_{\alpha=1}^k z_{f_\alpha} (a_\alpha I_{2m} + X_\alpha) \right).$$

Multiplying the contributions from the $\mathfrak{f}$ -block and the $\mathfrak{n}$ -block establishes the lemma. $\square$

Using the symmetry of the Heisenberg nilradical, we obtain a factorization of the characteristic polynomial:

Theorem 3.3. *Let $\mathfrak{s}$ be a solvable Lie algebra with Heisenberg nilradical $\mathfrak{n} \cong \mathfrak{h}(m)$ and generic derivation $D(z) = \sum z_\alpha \text{ad}(f_\alpha)|_{\mathfrak{n}}$. The characteristic polynomial admits the unique canonical factorization:*

$$Q_{\mathfrak{s}}(z_0, z) = z_0^k \cdot (z_0 + \gamma(z)) \cdot \prod_{i=1}^m \left(z_0 + \frac{\gamma(z)}{2} - \delta_i(z) \right) \left(z_0 + \frac{\gamma(z)}{2} + \delta_i(z) \right)$$

where $\gamma(z) = \frac{1}{m+1} \text{Tr } D(z)$ is the central weight, and $\{\pm\delta_i(z)\}_{i=1}^m$ are the spectrum of the centered derivation $D(z) - \frac{1}{2}\gamma(z)I$ acting on the quotient $\mathfrak{n}/Z(\mathfrak{n})$.

Proof. We use the explicit determinant expression obtained in Lemma 3.2. The factor z_0^k is immediate.

The linear factor arises from the action on the one-dimensional center $Z(\mathfrak{n})$. Writing $\gamma(z) = \sum 2a_\alpha z_\alpha$, we verify the trace formula by computing $\text{Tr } D(z)$. The restriction of $D(z)$ to the symplectic complement acts by $\sum z_\alpha (a_\alpha I + X_\alpha)$. Since each $X_\alpha \in \mathfrak{sp}(2m, \mathbb{C})$ is traceless, this block contributes $2m \sum a_\alpha z_\alpha$ to the trace. Adding the central contribution $\sum 2a_\alpha z_\alpha$ gives $\text{Tr } D(z) = (m+1)\gamma(z)$.

Finally, consider the determinant corresponding to the action on the quotient $V = \mathfrak{n}/Z(\mathfrak{n})$. This block has the form $M(z) = a(z)I_{2m} + S(z)$, where $a(z) = \gamma(z)/2$ and $S(z) = \sum z_\alpha X_\alpha \in \mathfrak{sp}(2m, \mathbb{C})$. The spectrum of $S(z)$ is symmetric about zero and consists of pairs $\pm\delta_i(z)$. Accordingly, the eigenvalues of $M(z)$ are $a(z) \pm \delta_i(z)$, and the determinant factors as

$$\det(z_0 I + M(z)) = \prod_{i=1}^m (z_0 + a(z) + \delta_i(z)) (z_0 + a(z) - \delta_i(z)).$$

Substituting $a(z) = \gamma(z)/2$ gives the stated factorization. The identification of $\{\pm\delta_i(z)\}$ with the spectrum of the centered derivation follows immediately. $\square$

This gives us a simpler formula for $k(\mathfrak{s})$:

Corollary 3.4. *For a solvable extension of $\mathfrak{h}(m)$, the spectral invariant $k(\mathfrak{s})$ is given by*

$$k(\mathfrak{s}) = \left| \{0, \gamma\} \cup \bigcup_{i=1}^m \left\{ \frac{\gamma}{2} - \delta_i, \frac{\gamma}{2} + \delta_i \right\} \right|,$$

where the linear forms are defined as in Theorem 3.3. In particular, we have the sharp bound $k(\mathfrak{s}) \leq 2m + 2$.

Now, we give a criterion that characterizes when a family of solvable Heisenberg extensions is spectrally equivalent.

Theorem 3.5. *Let $\mathfrak{s}$ and $\mathfrak{s}'$ be solvable Lie algebras with Heisenberg nilradical $\mathfrak{h}(m)$. Let γ, γ' denote their respective central weights, and let $\Delta = \{\delta_1, \dots, \delta_m\}$ and $\Delta' = \{\delta'_1, \dots, \delta'_m\}$ denote their respective multisets of shifts. The algebras $\mathfrak{s}$ and $\mathfrak{s}'$ are spectrally equivalent if and only if there exists a linear isomorphism $B : \mathfrak{f}' \rightarrow \mathfrak{f}$ such that:*

$$\gamma'(\mathbf{z}) = \gamma(\mathbf{z}B), \tag{1}$$

$$\{\delta'_j(\mathbf{z})^2\}_{j=1}^m = \{\delta_i(\mathbf{z}B)^2\}_{i=1}^m, \quad (2)$$

where the second equality holds as multisets of quadratic forms on $\mathfrak{f}'$.

Proof. For the forward implication, assume $\mathfrak{s}$ and $\mathfrak{s}'$ are spectrally equivalent. By definition, there exists an invertible linear map B such that $Q_{\mathfrak{s}'}(z_0, \mathbf{z}) = Q_{\mathfrak{s}}(z_0, \mathbf{z}B)$.

We first identify the central weights by inspecting the coefficients of the polynomial in the variable z_0 . Expanding $Q_{\mathfrak{s}}(z_0, \mathbf{z})$ in descending powers of z_0 , the second highest term is given by the negative sum of the roots. Using the canonical factorization from Theorem 3.3, the sum of the roots is:

$$\text{Tr}(\text{ad}_{\mathbf{z}}) = \gamma(\mathbf{z}) + \sum_{i=1}^m \left[\left(\frac{\gamma(\mathbf{z})}{2} - \delta_i(\mathbf{z}) \right) + \left(\frac{\gamma(\mathbf{z})}{2} + \delta_i(\mathbf{z}) \right) \right] = (m+1)\gamma(\mathbf{z}).$$

Since the polynomials are identical under the change of variables, the coefficients of z_0^{n-1} must match:

$$(m+1)\gamma'(\mathbf{z}) = (m+1)\gamma(\mathbf{z}B).$$

This implies $\gamma'(\mathbf{z}) = \gamma(\mathbf{z}B)$, establishing the first condition.

Next, we divide both sides of the equation $Q_{\mathfrak{s}'}(z_0, \mathbf{z}) = Q_{\mathfrak{s}}(z_0, \mathbf{z}B)$ by the common factor $z_0^k(z_0 + \gamma'(\mathbf{z}))$. This yields the equality of the remaining symplectic factors:

$$\prod_{j=1}^m \left[\left(z_0 + \frac{\gamma'(\mathbf{z})}{2} \right)^2 - \delta'_j(\mathbf{z})^2 \right] = \prod_{i=1}^m \left[\left(z_0 + \frac{\gamma(\mathbf{z})}{2} \right)^2 - \delta_i(\mathbf{z}B)^2 \right].$$

By the unique factorization of polynomials, the multisets of the quadratic factors must be identical. Since the term $(z_0 + \frac{1}{2}\gamma'(\mathbf{z}))^2$ is common to all factors on both sides, the equality holds if and only if the multisets of the constant terms (relative to z_0) are identical. Thus,

$$\{\delta'_j(\mathbf{z})^2\}_{j=1}^m = \{\delta_i(\mathbf{z}B)^2\}_{i=1}^m.$$

Conversely, suppose such a map B exists. The condition on γ ensures the linear factors $z_0 + \gamma$ align. The condition on the multisets of squared shifts implies that the product of the quadratic factors in $Q_{\mathfrak{s}'}(z_0, \mathbf{z})$ is identical to that in $Q_{\mathfrak{s}}(z_0, \mathbf{z}B)$. Combining these finishes the proof. $\square$

We can also specialize Corollary 2.7 to the Heisenberg nilradical setting.

Theorem 3.6. *Let $\mathfrak{s}(\mathbf{p})$ be a family of solvable Heisenberg extensions with central weight γ and shifts $\{\delta_1, \dots, \delta_m\}$. Suppose δ_i admits a decomposition*

$$\delta_i(z; \mathbf{p}) = \mu(\mathbf{p})\gamma(z) + \tau(z),$$

where τ is a fixed linear form. If τ is linearly independent of the subspace spanned by γ and the remaining shifts $\{\delta_j\}_{j \neq i}$, then the family $\mathfrak{s}(\mathbf{p})$ is not spectrally rigid.

Proof. Let $\mathbf{p}$ and $\mathbf{p}'$ be parameters differing only in μ , with difference $\Delta\mu = \mu(\mathbf{p}') - \mu(\mathbf{p})$. According to the canonical factorization of the Heisenberg characteristic polynomial, the multiset of weights is given by:

$$\Delta(\mathbf{p}) = \{0, \gamma\} \cup \bigcup_{k=1}^m \left\{ \frac{\gamma}{2} - \delta_k(\mathbf{p}), \frac{\gamma}{2} + \delta_k(\mathbf{p}) \right\}.$$

To apply Corollary 2.7, we must demonstrate the existence of an automorphism $T \in \text{GL}(\mathfrak{f}^*)$ satisfying $T(\Delta(\mathbf{p})) = \Delta(\mathbf{p}')$.

By hypothesis, the set $S = \{\gamma, \tau\} \cup \{\delta_j\}_{j \neq i}$ is linearly independent. We extend S to a basis of $\mathfrak{f}^*$ and define T as the unique linear map satisfying:

$$\begin{aligned} T(\gamma) &= \gamma, \\ T(\tau) &= \tau + \Delta\mu \gamma, \\ T(\delta_j) &= \delta_j \quad \text{for all } j \neq i, \end{aligned}$$

fixing any remaining basis elements.

We now verify the action of T on the weights involving the i -th shift. Using the linearity of T and the decomposition of δ_i :

$$\begin{aligned}
T(\delta_i(\mathbf{p})) &= T(\mu(\mathbf{p})\gamma + \tau) \\
&= \mu(\mathbf{p})T(\gamma) + T(\tau) \\
&= \mu(\mathbf{p})\gamma + (\tau + \Delta\mu\gamma) \\
&= (\mu(\mathbf{p}) + \Delta\mu)\gamma + \tau \\
&= \mu(\mathbf{p}')\gamma + \tau \\
&= \delta_i(\mathbf{p}').
\end{aligned}$$

Since T fixes γ and δ_j for $j \neq i$, it maps the subsets of weights $\{\frac{\gamma}{2} \pm \delta_k(\mathbf{p})\}$ to $\{\frac{\gamma}{2} \pm \delta_k(\mathbf{p}')\}$ for all k . Consequently, T maps the entire multiset $\Delta(\mathbf{p})$ bijectively onto $\Delta(\mathbf{p}')$. By Corollary 2.7, the algebras are spectrally equivalent. $\square$

3.2 Topological Spectral Invariants

The symplectic structure of the Heisenberg nilradical imposes strict dimension constraints on the cohomology of the eigenvariety complement, strengthening the general bounds from Section 2.

Theorem 3.7. *Let $\mathfrak{s}$ be a solvable Lie algebra with Heisenberg nilradical $\mathfrak{h}(m)$. Then the Betti numbers satisfy $b_j(\mathfrak{s}) = 0$ for all $j > m + 1$.*

Proof. By Theorem 2.22, the degree of the Poincaré polynomial equals the dimension of the subspace of $\mathfrak{f}^*$ spanned by the nilradical weights Δ . According to the canonical factorization in Theorem 3.3, the set of weights is

$$\Delta = \{0, \gamma\} \cup \bigcup_{i=1}^m \left\{ \frac{\gamma}{2} - \delta_i, \frac{\gamma}{2} + \delta_i \right\}.$$

It is immediate that $\text{span}(\Delta) \subseteq \text{span}\{\gamma, \delta_1, \dots, \delta_m\}$. Since this subspace has dimension at most $m + 1$, the result follows. $\square$

Next, we strengthen Corollary 2.24. The general bound assumes the linear forms are in general position, but the symplectic structure enforces specific linear dependencies among the defining forms, preventing the arrangement from being in general position. Recall that for a matroid $\mathcal{M}$ defined on a ground set Φ , a subset $S \subseteq \Phi$ is called a circuit if it is a minimally dependent set, such that every proper subset of S is independent. The rank function $\text{rank} : 2^\Phi \rightarrow \mathbb{Z}_{\geq 0}$ of the matroid is defined such that for any subset $A \subseteq \Phi$, $\text{rank}(A)$ is the maximum cardinality of an independent set contained in A . In the context of the spectral matroid $\mathcal{M}_s$, the rank of a set of linear forms is equivalent to the dimension of their span in the dual space $(\mathbb{C} \oplus \mathfrak{f})^*$.

Lemma 3.8. *Let Φ be the set of linear forms defining the hyperplane arrangement $\mathcal{A}_s$. Suppose $\gamma \neq 0$ and $\delta_i \notin \{0, \pm\frac{\gamma}{2}\}$. For each $i \in \{1, \dots, m\}$, the subset*

$$C_i = \left\{ z_0, z_0 + \gamma, z_0 + \frac{\gamma}{2} - \delta_i, z_0 + \frac{\gamma}{2} + \delta_i \right\} \subset \Phi$$

is linearly dependent with rank 3, forming a circuit of size 4 in the spectral matroid $\mathcal{M}_s$.

Proof. We explicitly construct the dependence relation. Let $\ell_1 = z_0$, $\ell_2 = z_0 + \gamma$, and define the symplectic forms $\ell_3 = z_0 + \frac{\gamma}{2} - \delta_i$ and $\ell_4 = z_0 + \frac{\gamma}{2} + \delta_i$. Consider the linear combination $\sum_{k=1}^4 c_k \ell_k$ with coefficients $c_1 = c_2 = 1$ and $c_3 = c_4 = -1$. Substituting the definitions gives:

$$(\ell_1 + \ell_2) - (\ell_3 + \ell_4) = (2z_0 + \gamma) - \left(\left(z_0 + \frac{\gamma}{2} - \delta_i \right) + \left(z_0 + \frac{\gamma}{2} + \delta_i \right) \right) = 0.$$

This proves that C_i is a dependent set. To determine the rank, observe that under the non-degeneracy assumptions, the vectors are pairwise linearly independent. Furthermore, since δ_i is generic relative to γ , no linear relation exists among any subset of size three (e.g., ℓ_1, ℓ_2, ℓ_3 are independent). Therefore, every proper subset of C_i is linearly independent, implying that C_i is a minimal dependent set (a circuit) of rank 3. $\square$

This dependency structure forces a strictly stronger bound on the higher Betti numbers via the log-concavity of the intersection lattice.

Theorem 3.9. *Let $n = k(\mathfrak{s})$ be the number of hyperplanes. For degrees $j \geq 3$, the Betti numbers satisfy*

$$b_j(\mathfrak{s}) \leq \binom{n}{j} \left(1 - \frac{6}{n(n-1)}\right)^{j-2}.$$

Proof. Consider the normalized Betti numbers $x_j = b_j(\mathfrak{s}) \binom{n}{j}^{-1}$. For $j \leq 2$, the genericity of the arrangement in low codimension implies $b_j(\mathfrak{s}) = \binom{n}{j}$, and thus $x_0 = x_1 = x_2 = 1$.

For $j = 3$, we utilize the isomorphism between the cohomology ring $H^*(\mathcal{M}(\mathcal{A}_{\mathfrak{s}}))$ and the Orlik-Solomon algebra E/I . The dimension of the third graded component is given by $\dim(E^3) - \dim(I^3)$. In a generic arrangement, $I^3 = 0$. However, Lemma 3.8 establishes that $\mathcal{M}_{\mathfrak{s}}$ contains circuits of size 4. For each such circuit C_i , the boundary element $\partial(e_{C_i})$ is a non-zero vector in the ideal I of degree 3. This strict increase in the dimension of the ideal implies $b_3(\mathfrak{s}) < \binom{n}{3}$, and therefore we have the bound:

$$x_3 \leq 1 - \frac{6}{n(n-1)}.$$

By Theorem 2.29, the sequence $\{x_j\}$ is log-concave, which implies that the sequence of ratios $r_j = x_j/x_{j-1}$ is non-increasing. From the initial values, we have $r_1 = r_2 = 1$. Consequently, for all $j \geq 3$, the ratio is bounded by the first non-trivial term: $r_j \leq r_3 = x_3$.

The result follows by induction on j . Writing $x_j = x_2 \prod_{k=3}^j r_k$, and substituting the bounds $x_2 = 1$ and $r_k \leq x_3$, we obtain:

$$x_j \leq 1 \cdot (x_3)^{j-2} = \left(1 - \frac{6}{n(n-1)}\right)^{j-2}.$$

Multiplying by the normalization factor $\binom{n}{j}$ gives the stated inequality. $\square$

3.3 Computations

Using the classification of [RW93], we compute the characteristic polynomial, the spectral invariant $k(\mathfrak{s})$, and the Poincaré polynomials $P_{\mathfrak{s}}(t)$ for all solvable Lie algebras with three- and five-dimensional Heisenberg nilradical.

Table 1: Solvable Lie Algebras with Heisenberg Nilradical

Dim	Algebra $\mathfrak{s}$	Characteristic Polynomial $Q_{\mathfrak{s}}(z)$	$k(\mathfrak{s})$	$P_{\mathfrak{s}}(t)$
4	$\mathfrak{s}_{3,1}^{0,1}$	$z_0^2(z_0 - z_4)(z_0 + z_4)$	3	$2t + 1$
4	$\mathfrak{s}_{3,1}^{0,2}$	$z_0(z_0 + z_4)^2(z_0 + 2z_4)$	3	$2t + 1$
4	$\mathfrak{s}_{3,1}^{1,1,b}$			
	$b \geq 0, b \neq 1$	$z_0(z_0 + 2z_4)(z_0 + (1-b)z_4)(z_0 + (1+b)z_4)$	4	$3t + 1$
	$b = 1$	$z_0^2(z_0 + 2z_4)^2$	2	$t + 1$
	$b = 0$	$z_0(z_0 + 2z_4)(z_0 + z_4)^2$	3	$2t + 1$
5	$\mathfrak{s}_{3,2}^{0,1}$	$z_0^2(z_0 + 2z_4)(z_0 + z_4 - z_5)(z_0 + z_4 + z_5)$	4	$3t^2 + 3t + 1$
6	$\mathfrak{s}_{5,1}^{0,1}$	$z_0(z_0 + z_6)^4(z_0 + 2z_6)$	3	$2t + 1$
6	$\mathfrak{s}_{5,1}^{0,2}$	$z_0^4(z_0 - z_6)(z_0 + z_6)$	3	$2t + 1$
6	$\mathfrak{s}_{5,1}^{0,3}$	$z_0^2(z_0 - z_6)^2(z_0 + z_6)^2$	3	$2t + 1$

Dim	Algebra $\mathfrak{s}$	$Q(z)$	$k(L)$	$P(t)$
6	$\mathfrak{s}_{5,1}^{0,4}$	$z_0(z_0 + z_6)^4(z_0 + 2z_6)$	3	$2t + 1$
6	$\mathfrak{s}_{5,1}^{1,1,c}$			
	$0 < c < 1$	$z_0^2(z_0 - z_6)(z_0 + z_6)(z_0 - cz_6)(z_0 + cz_6)$	5	$4t + 1$
	$c = 1$	$z_0^2(z_0 - z_6)^2(z_0 + z_6)^2$	3	$2t + 1$
6	$\mathfrak{s}_{5,1}^{1,2,b}$			
	$0 \leq \arg b \leq \pi$	$z_0(z_0 + z_6)^2(z_0 + 2z_6)(z_0 + (1 - b)z_6)(z_0 + (1 + b)z_6)$	5	$4t + 1$
	$b = 1$	$z_0(z_0 + z_6)^3(z_0 + 2z_6)^2$	3	$2t + 1$
6	$\mathfrak{s}_{5,1}^{1,3,b}$			
	$0 \leq \arg b \leq \pi$	$z_0(z_0 + 2z_6)(z_0 + (1 - b)z_6)^2(z_0 + (1 + b)z_6)^2$	4	$3t + 1$
	$b = 1$	$z_0^3(z_0 + 2z_6)^3$	2	$t + 1$
6	$\mathfrak{s}_{5,1}^{2,1,b,c}$			
	$ c \leq b $ $0 \leq \arg b, c \leq \pi$	$z_0(z_0 + 2z_6)(z_0 + (1 - b)z_6)(z_0 + (1 + b)z_6)(z_0 + (1 - c)z_6)(z_0 + (1 + c)z_6)$	6	$5t + 1$
	$b = 1$ (requires $ c < 1$)	$z_0^3(z_0 + 2z_6)^3(z_0 + (1 - c)z_6)(z_0 + (1 + c)z_6)$	4	$3t + 1$
7	$\mathfrak{s}_{5,2}^{0,1}$	$z_0^3(z_0 - z_6)(z_0 + z_6)(z_0 - z_7)(z_0 + z_7)$	5	$4t^2 + 4t + 1$
7	$\mathfrak{s}_{5,2}^{0,2}$	$z_0^3(z_0 - z_6)(z_0 + z_6)(z_0 - z_7)(z_0 + z_7)$	5	$4t^2 + 4t + 1$
7	$\mathfrak{s}_{5,2}^{0,3}$	$z_0^4(z_0 + 2z_7)^3$	2	$t + 1$
7	$\mathfrak{s}_{5,2}^{0,4}$	$z_0^4(z_0 + 2z_7)^3$	2	$t + 1$
7	$\mathfrak{s}_{5,2}^{0,5}$	$z_0^3(z_0 + z_7)^2(z_0 + 2z_7)^2$	3	$2t + 1$
7	$\mathfrak{s}_{5,2}^{1,1,b}$			
	$0 \leq \arg b < \pi$	$z_0^2(z_0 + z_7)^2(z_0 + 2z_7)(z_0 + bz_6)(z_0 - bz_6 + 2z_7)$	5	$4t^2 + 4t + 1$
7	$\mathfrak{s}_{5,2}^{1,2,b}$			
	$0 \leq \arg b < \pi$	$z_0^3(z_0 - bz_6 + z_7)^2(z_0 + bz_6 + z_7)^2$	3	$2t + 1$
7	$\mathfrak{s}_{5,2}^{2,1,b,c}$			
	$0 < \arg b < \pi$ $ c \leq 1$	$z_0^3(z_0 + 2z_7)^2(z_0 - bz_6 + (1 - c)z_7)(z_0 + bz_6 + (1 + c)z_7)$	4	$3t^2 + 3t + 1$
8	$\mathfrak{s}_{5,3}^{0,1}$	$z_0^3(z_0 + 2z_6)(z_0 + z_6 - z_7)(z_0 + z_6 + z_7)$ $(z_0 + z_6 - z_8)(z_0 + z_6 + z_8)$	6	$7t^3 + 10t^2 + 5t + 1$

3.4 Spectral Equivalence Classification

Using the computations in the previous subsection, we can now give a complete spectral classification for all solvable Lie algebras with three- and five-dimensional Heisenberg nilradical.

Table 2: Spectral Classification

Dim	Algebra	Parameters	Spectral Equivalences
4	$\mathfrak{s}_{3,1}^{0,1}$	—	—
	$\mathfrak{s}_{3,1}^{1,1}(b)$	$b \geq 0$	$\mathfrak{s}_{3,1}^{1,1}(0) \sim \mathfrak{s}_{3,1}^{0,2}$
5	$\mathfrak{s}_{3,2}^{0,1}$	—	—
6	$\mathfrak{s}_{5,1}^{1,1}(c)$	$0 \leq c \leq 1$	$\mathfrak{s}_{5,1}^{1,1}(0) \sim \mathfrak{s}_{5,1}^{0,2}$ $\mathfrak{s}_{5,1}^{1,1}(1) \sim \mathfrak{s}_{5,1}^{0,3}$
	$\mathfrak{s}_{5,1}^{2,1}(b, c)$	$ c \leq b $	$\mathfrak{s}_{5,1}^{2,1}(0, 0) \sim \mathfrak{s}_{5,1}^{0,1} \sim \mathfrak{s}_{5,1}^{0,4}$
		$0 \leq \arg(b, c) \leq \pi$	$\mathfrak{s}_{5,1}^{2,1}(1, 0) \sim \mathfrak{s}_{5,1}^{1,2}$
			$\mathfrak{s}_{5,1}^{2,1}(1, 1) \sim \mathfrak{s}_{5,1}^{1,3}$
7	$\mathfrak{s}_{5,2}^{0,1}$	—	$\mathfrak{s}_{5,2}^{0,1} \sim \mathfrak{s}_{5,2}^{0,2}$
	$\mathfrak{s}_{5,2}^{0,3}$	—	$\mathfrak{s}_{5,2}^{0,3} \sim \mathfrak{s}_{5,2}^{0,4}$
	$\mathfrak{s}_{5,2}^{1,1}(b)$	$0 \leq \arg b < \pi$	$\mathfrak{s}_{5,2}^{1,1}(0) \sim \mathfrak{s}_{5,2}^{0,5}$
			$\forall b \neq 0 : \mathfrak{s}_{5,2}^{1,1}(b) \sim \mathfrak{s}_{5,2}^{1,1}(1)$
	$\mathfrak{s}_{5,2}^{1,2}(b)$	$0 \leq \arg b < \pi$	$\forall b \neq 0 : \mathfrak{s}_{5,2}^{1,2}(b) \sim \mathfrak{s}_{5,2}^{1,2}(1)$
	$\mathfrak{s}_{5,2}^{2,1}(b, c)$	$ c \leq 1$	—
		$0 < \arg b < \pi$	
8	$\mathfrak{s}_{5,3}^{0,1}$	—	—

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